# Nathan Stangler

Minneapolis, MN
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https://nathanstangler.github.io

## EDUCATION

**Master of Science in Computer Science** Expected May 2027
University of Minnesota-Twin Cities Minneapolis, MN
Minor: Cyber Security

**Bachelor of Science** May 2026
University of Minnesota-Twin Cities Minneapolis, MN
Majors: Computer Science, Data Science; Minor: Statistics
GPA: 3.976/4.00

## PROFESSIONAL EXPERIENCE

**Amazon Web Services** May 2026-August 2026
*Software Development Engineer Intern* Minneapolis, MN

- Engineered a challenge-response authentication system for secure serial access to Linux networking devices, implementing a Rust PAM module to validate cryptographically signed, user-specific credentials.
- Built a Go-based signing service to authorize device access and issue signed credentials using a certificate chain, enabling offline credential validation and user-level auditing.
- Developed a React Web Serial application with an integrated terminal to automate device authentication and credential exchange, simplifying secure access for data center operators.

**ImageTrend** May 2025-August 2025
*Software Developer Intern* Eagan, MN

- Designed and implemented a full-stack geocoding data visibility and control application with a Windows service API using C# and .NET Core.
- Created an interface for managing geocoding data sources and viewing logs using React.

**Best Buy** July 2021-April 2023
*Sales Advisor* Maple Grove, MN

- Trained new employees in technical aspects of mobile device activation and setup.
- Provided technical support for customers by using critical thinking skills to ease concerns.

## RESEARCH EXPERIENCE

**University of Minnesota, Knowledge Computing Lab** May 2025-May 2026
*Undergraduate Researcher (Mentor: JangHyeon Lee, Advisor: Yao-Yi Chiang)* Minneapolis, MN

- Led development of SolarCLIP, a domain-adapted vision–language model for solar physics by selectively fine-tuning CLIP's projection heads and targeted transformer layers.
- Designed and implemented a full machine learning experimentation pipeline including model training, evaluation tooling, data preprocessing, and embedding-space analysis.

## TEACHING EXPERIENCE

**University of Minnesota, Department of Computer Science & Engineering** Jan 2026-May 2026
*Undergraduate Teaching Assistant (CSCI 5541: Natural Language Processing)* Minneapolis, MN

- Supported instruction for a graduate-level NLP course covering word embeddings, transformers, and large language models.
- Mentored students through semester-long projects involving dataset collection, annotation, model design, training, and evaluation methodologies.
- Provided detailed grading and technical feedback on assignments and projects.

## SELECTED PROJECTS

**RelayMQ** Fall 2026

- Built a C++ publish/subscribe message broker with non-blocking network I/O, wildcard subject routing, token-based access control, and support for standalone and peer-to-peer clustered deployments.
- Developed C++ and Go client SDKs and a CLI, enabling applications to connect to RelayMQ, publish messages, and subscribe to subjects.

**API Monitor** Summer 2026

- Developed a distributed API monitoring platform with scheduled polling, NATS, PostgreSQL, and worker services to continuously track endpoint uptime, health, and latency.
- Built a real-time dashboard with Google OAuth, configurable response schemas, and WebSockets to detect unreachable endpoints and breaking API changes, with public tracker discovery for shared monitoring.

**Truncate** Summer 2025

- Built a full-stack URL shortener and analytics platform using React, TypeScript, Go (Fiber), and PostgreSQL.
- Implemented detailed analytics, including redirect tracking, geolocation, device, and browser statistics.
- Secured user data with JWT authentication and role-based access control.

**Improved Scoreboard** Summer 2024

- Designed and developed a scalable, object oriented, and event driven Minecraft network packet-based scoreboard plugin compatible with multiple game versions.
- Integrated with third-party applications through the creation of a portable scoreboard management API.

**Daily Rewards** Fall 2020

- Developed and commercially launched a fully configurable daily rewards system for Garry's Mod, generating 100+ sales on Gmodstore.
- Integrated MySQL and SQLite database support, with seamless interaction from a custom designed user interface using client-server communication through networking protocols.

## ACTIVITIES

**Member,** University of Minnesota Robotics Club September 2023-May 2024

- Contributed to the creation and design of an autonomous robot for the NASA Lunabotics competition using NVIDIA Isaac ROS libraries.
- Collaborated with club members to develop and debug a heightmap conveyor belt detector and a vision-based position estimator.

## AWARDS AND HONORS

Dean's List, University of Minnesota, Minneapolis, MN, 2023-2026

## SKILLS

**Languages:** Go, Java, C++, C, Python, C#, SQL, JavaScript/TypeScript
**Libraries:** PyTorch, NumPy, Pandas, Scikit-Learn, React
**Tools:** Git, GitHub, HuggingFace, PostgreSQL